# Feiyang **Chen**

AI Application & Systems Software Engineer

*Suzhou, Jiangsu, China*

(+86) 188-5297-8070 | chris.chenfeiyang@gmail.com | GitHub / Open Source FreeFlyingSheep | Technical Blog freeflyingsheep.github.io

## Summary

University of Sydney computer science master's graduate with approximately three years of Linux kernel and systems software development experience. Independently developed and maintain a news proofreading agent for in-house reporters; contributed to a content management platform. Skilled in OS porting, cross-architecture adaptation, and software debugging; experienced in AI applications and low-level systems software.

## Skills

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| **Languages** | Python, C |
| **AI Applications** | LangChain/LangGraph, RAG, MCP |
| **Backend** | FastAPI, PostgreSQL, Redis |
| **Engineering** | Linux development, cross-architecture porting, Docker |

## Projects

### News Proofreading and Content Management

*Suzhou, Jiangsu, China*

Project Collaborator | Suzhou Xinsu Baoye Network Technology Co., Ltd. — *Apr. 2026 – Present*

- **AI Proofreading:** Independently developed and maintain a FastAPI/LangChain service for language and factual-consistency checks; used by the group's in-house reporters, averaging 100+ calls per day.
- **Fact Retrieval:** Built news ingestion, entity/appointment extraction, and PostgreSQL retrieval, using alias matching and evidence update times to provide traceable evidence for time-sensitive news checks.
- **Output Validation:** Validated suggestion formats and source locations with Pydantic and unique source-span checks; fed errors back for retries and capped model/tool calls.
- **Editor and MCP:** Built an SDK for accepting/rejecting suggestions and applying edits while preserving Markdown/HTML; integrated WorkBuddy through FastMCP with JWT authentication, rate limits, quota consumption, and refunds for pre-execution failures.
- **Runtime and Testing:** Deployed Qwen locally; implemented asynchronous ingestion, request deduplication, and result caching with Taskiq/Redis; added Langfuse tracing and pytest/Testcontainers and Vitest tests for retrieval, validation, deduplication, and edit application.
- **Business System:** Developed the content management frontend/backend and integrated the portal within a team, covering multi-tenancy, legacy data migration, third-party synchronization, and publishing schedules.
- **Personal Technical Validation:** Validated LangGraph routing/query rewriting and BM25/vector retrieval with RRF in a separate open-source A-share analysis agent; evaluated faithfulness and tool use with DeepEval and documented architecture choices and lessons learned.

### Systems Software and Open-source Contributions

*Suzhou, Jiangsu, China*

Kernel and Systems Development (Including Open-source Maintenance) — *Jun. 2020 – Feb. 2025*

- Maintained a separate LoongArch64/Linux branch of Valgrind; independently implemented libVEX support for the base instruction set and adapted the runtime, and contributed to vector instruction support; reached initial usability within one month, passed 100+ regression tests, and added platform/nightly CI test configurations, advancing low-level toolchain compatibility with LoongArch.
- Contributed to Linux system call and network driver porting; added LoongArch32/64 support to nolibc, statx compatibility, and platform selftests, merged into mainline Linux; added LoongArch support to Kcov, merged upstream.
- In separately maintained branches, added LoongArch/QEMU support to RT-Thread, contributed to LS2K0300 integration, and ported Newlib.

## Work Experience

### Loongson Technology Corporation Limited

*Suzhou, Jiangsu, China*

Kernel Engineer — *Dec. 2020 – Aug. 2023*

- Contributed to MIPS/LoongArch Linux kernel development and upstream patch submissions, most merged into mainline; diagnosed defects, implemented fixes, and ran regression tests.
- Led LoongArch porting for Valgrind and other foundational software, resolving cross-architecture build, instruction compatibility, runtime, and performance issues.
- Implemented CI/CD for the company's documentation repository, automating document builds and preview website generation so community members could preview changes and collaborate on documentation.

### Jiangsu Lemote Technology Co., Ltd.

*Suzhou, Jiangsu, China*

Kernel Engineer — *Jun. 2020 – Dec. 2020*

- Developed and debugged Linux kernel code through issue reproduction, log analysis, patching, and validation.
- Ported OpenStack Nova and DevStack to Fedora MIPS in separately maintained branches, resolving component compatibility and environment issues; brought up the core service workflow within one month.

## Education

### The University of Sydney

*Sydney, NSW, Australia*

Master of Computer Science — *Feb. 2024 - May 2026*

- **ISSRE 2026 accepted paper (second author):** "Dr. DD: 1-Minimal Isolation of Failure Causes via Deferred Restarts."
- **Implementation and Experiments:** Built a Python delta debugging framework to automatically reduce failure-inducing inputs for binary software, aiding bug reproduction and diagnosis; validated it on Binutils/Valgrind, comparing reduction quality, test calls, and execution time.

### Suzhou University of Technology (formerly Changshu Institute of Technology)

*Suzhou, Jiangsu, China*

Bachelor of Computer Science and Technology (Industry-integrated Program) — *Sep. 2016 - Jun. 2020*

- **ICAIS 2020 paper (first author):** "Formal Verification of the Correctness of Operating System on Assembly Layer in Isabelle/HOL."
- **Research:** Modelled MINIX 3's i386 context-saving assembly in Isabelle/HOL and verified register/memory states and instruction execution against expected results.